

Mutual linearity from classical response theory

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We unify known mutual-linearity relations for Markov jump processes and chemical reaction networks within a common framework based on classical finite-response theory. In a Markov jump process, an exact finite-response identity shows that all stationary probability changes share a common factor when the rates of one edge change. Eliminating this factor gives the relations between probabilities, observables, and currents, even far from equilibrium. We also explain how the same argument applies when one coefficient of a linear system changes and recovers the known relations in Laplace space and between stationary complex activities in chemical reaction networks.

I. INTRODUCTION

Stationary probabilities and currents generally depend nonlinearly on transition rates. Nevertheless, varying the rates of one edge produces straight-line relations between these quantities, known as mutual linearity [1–7]. Our aim is to clarify the common origin of these relations and provide a concise derivation using the Drazin inverse. We start from the classical finite-response identity [8, 9]. When the rates of one edge change, this identity shows that all stationary probability changes share a common factor. Eliminating it gives the relations between probabilities, observables, and currents. This approach gives short derivations of the stationary and Laplace-space relations [1, 6, 7] and the relations between stationary complex activities in chemical reaction networks [10]. We also explain when the same property holds in other linear systems.

We focus on the common derivation of these relations. The works cited above also establish more detailed results beyond the scope of this paper. These include explicit network formulas for the coefficients relating currents [1], expressions for these coefficients in terms of mean first-passage times [2], relations for several input currents [3], bounds on stationary probabilities and observables [5], and relations between Laplace transforms of first-hitting-time distributions [7].

II. EXACT STATIONARY RESPONSE

The finite-response identity shown below dates back to the 1960s [8] and was originally formulated for Markov chains. Here we write it for continuous-time Markov jump processes on a finite state space.

We consider a finite irreducible Markov jump process, $\partial_t \mathbf{p} = W \mathbf{p}$, where W_{ij} is the transition rate from j to i , and $W_{jj} = -\sum_{i \neq j} W_{ij}$. The stationary distribution satisfies $W \boldsymbol{\pi} = 0$ and $\mathbf{1}^\top \boldsymbol{\pi} = 1$. For a second irreducible genera-

tor $W' = W + \Delta W$, the classical fundamental-matrix identity takes the form

$$\Delta \boldsymbol{\pi} \equiv \boldsymbol{\pi}' - \boldsymbol{\pi} = -G \Delta W \boldsymbol{\pi}', \quad (1)$$

where $G = W^D$ is the Drazin inverse [11–13]. It satisfies $GW = WG = I - \boldsymbol{\pi} \mathbf{1}^\top$, $G \boldsymbol{\pi} = 0$, and $\mathbf{1}^\top G = 0$. Indeed, subtracting the two stationary equations gives $W \Delta \boldsymbol{\pi} = -\Delta W \boldsymbol{\pi}'$. Multiplication by G and normalization give Eq. (1), without assuming that ΔW is small.

The Drazin inverse underlies current-cumulant calculations [14], expansions under slow driving [15], fluctuation-response relations for state observables [16] and state-current correlations [17], responses in the Curie–Weiss model [18], and finite perturbations of Markov jump processes [19]. The relation $\tau_{j \rightarrow i} = (G_{ij} - G_{ii})/\pi_i$ connects this approach to mean first-passage times [2, 19–21]. Alternatively, normalization gives an invertible matrix from the rate matrix [22, 23].

III. MUTUAL LINEARITY

A. State observables

We start with the derivation of mutual linearity for probabilities and state observables. For the second matrix W' , we change W_{ij} and W_{ji} , keeping all other non-diagonal rates fixed and adjusting the diagonal elements to preserve probability conservation. Then, only components i and j of the vector $\Delta W \boldsymbol{\pi}'$ can be nonzero, and they sum to zero. Equation (1) therefore gives

$$\begin{aligned} \Delta \pi_n &= (G_{nj} - G_{ni})(\Delta W_{ij} \pi'_j - \Delta W_{ji} \pi'_i) \\ &\equiv h_n \alpha, \end{aligned} \quad (2)$$

where $h_n \equiv G_{nj} - G_{ni}$ is calculated from the rates before the change and does not depend on ΔW_{ij} or ΔW_{ji} . All dependence on these rate changes enters the second factor $\alpha \equiv \Delta W_{ij} \pi'_j - \Delta W_{ji} \pi'_i$. Repeating Eq. (2) for state m and eliminating α gives

$$\Delta \pi_n = \frac{h_n}{h_m} \Delta \pi_m, \quad h_m \neq 0. \quad (3)$$

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Thus, (π'_m, π'_n) lies on one straight line for every choice of the new edge rates. This property was recently discovered in Ref. [5].

More generally, for state observables $O = \sum_n O_n \pi_n$ and $Q = \sum_n Q_n \pi_n$ with fixed weights, the same argument gives

$$\Delta O = \frac{\sum_n O_n h_n}{\sum_n Q_n h_n} \Delta Q, \quad (4)$$

provided that the denominator is nonzero.

Let us now consider the linear-response limit of Eq. (2):

$$\partial_{W_{ij}} \pi_n = h_n \pi_j, \quad \partial_{W_{ji}} \pi_n = -h_n \pi_i, \quad (5)$$

so that $-\partial_{W_{ij}} \pi_n / \partial_{W_{ji}} \pi_n = \pi_j / \pi_i$ whenever $h_n \neq 0$. This recovers the response ratio of Ref. [5]. Interestingly, our dynamical fluctuation-response theory [24] gives the same ratio for dynamical responses to impulsive rate perturbations at time t , with π_j / π_i replaced by $p_j(t) / p_i(t)$. Thus, this response ratio can be obtained without invoking mutual linearity under finite rate changes.

B. Currents

Now we turn to mutual linearity between the steady-state currents. The stationary current from l to k is $J_{kl} \equiv W_{kl} \pi_l - W_{lk} \pi_k$. For an edge whose transition rates W_{kl} and W_{lk} are kept fixed, Eq. (2) gives

$$\Delta J_{kl} = (W_{kl} h_l - W_{lk} h_k) \alpha, \quad (6)$$

where only the change in the stationary distribution contributes. On the perturbed edge, the rates themselves also change:

$$\begin{aligned} \Delta J_{ij} &= W_{ij} \Delta \pi_j - W_{ji} \Delta \pi_i + \Delta W_{ij} \pi'_j - \Delta W_{ji} \pi'_i \\ &= (1 + W_{ij} h_j - W_{ji} h_i) \alpha. \end{aligned} \quad (7)$$

Eliminating α between the currents on (i, j) and another edge (k, l) gives

$$\Delta J_{kl} = \frac{W_{kl} h_l - W_{lk} h_k}{1 + W_{ij} h_j - W_{ji} h_i} \Delta J_{ij}, \quad (8)$$

which is the mutual linearity of stationary currents [1–4]. Both current changes contain the same α , so the coefficient relating them is independent of the new edge rates, provided the denominator in Eq. (8) is nonzero.

IV. FURTHER APPLICATIONS

We now apply the common-factor argument to Laplace-transformed dynamics and stationary complex activities. First, consider a linear system $A\mathbf{x} = \mathbf{b}$ with nonsingular A and fixed $\mathbf{b}$. If only A_{ij} changes and A' is nonsingular, subtracting the equations and eliminating $\Delta A_{ij} x'_j$ between n and m gives

$$\begin{aligned} \Delta x_n &= -(A^{-1})_{ni} \Delta A_{ij} x'_j, \\ \Delta x_n &= \frac{(A^{-1})_{ni}}{(A^{-1})_{mi}} \Delta x_m, \quad (A^{-1})_{mi} \neq 0. \end{aligned} \quad (9)$$

The same proportionality holds for fixed linear observables. Moreover, this argument applies whenever $\Delta A\mathbf{x}' = \mathbf{u}\alpha$, with fixed $\mathbf{u}$: the perturbation enters the equations in fixed proportions, with only its scalar amplitude α changing. Then $\Delta \mathbf{x} = -A^{-1} \mathbf{u} \alpha$, so eliminating α again gives mutual linearity.

A. Laplace-transformed dynamics

The frequency-domain extension follows from this argument using $A = sI - W$. For time-independent rates, the Laplace-transformed master equation is $(sI - W)\tilde{\mathbf{p}}(s) = \mathbf{p}(0)$. We keep the same initial distribution in both systems and define the resolvent matrix $R(s) = (sI - W)^{-1}$ for $\text{Re } s > 0$. Changing the rates of edge (i, j) and eliminating the common factor between states n and m results in

$$\begin{aligned} \Delta \tilde{p}_n &= h_n(s) (\Delta W_{ij} \tilde{p}'_j - \Delta W_{ji} \tilde{p}'_i), \\ \Delta \tilde{p}_n &= \frac{h_n(s)}{h_m(s)} \Delta \tilde{p}_m, \quad h_m(s) \neq 0, \end{aligned} \quad (10)$$

where $h_n(s) = R_{ni}(s) - R_{nj}(s)$ and all transformed probabilities are evaluated at the same s . This recovers the Laplace-domain probability relations [6, 7]. Repeating Eqs. (6)–(8) with $\tilde{\mathbf{p}}(s)$ and $h_n(s)$ gives the corresponding current relations [1]. These relations hold for the Laplace transforms. At a fixed time, varying the edge rates need not produce a straight-line relation between two probabilities or two currents [7].

B. Chemical reaction networks

The same argument applies to chemical reaction networks and recovers the mutual linearity of complex activities derived in Ref. [10]. Here we obtain this relation directly using the Drazin inverse. A *complex* is a combination of species appearing on either side of a reaction. For example, $A + B \rightleftharpoons C$ connects the complexes $A + B$ and C . If x_A, x_B, x_C are the species concentrations, the corresponding *complex activities* are $\psi_{A+B} = x_A x_B$ and $\psi_C = x_C$.

We represent the complexes as graph vertices and the reactions as edges, and consider a connected mass-action network with bidirectional reactions. The matrix K plays the same role as W above: for $i \neq j$, K_{ij} is the rate constant for the reaction $j \rightarrow i$, and each column sums to zero. We assume that the positive stationary states considered below are complex balanced [25]: at each complex, the total incoming reaction flux equals the total outgoing flux, so $K\boldsymbol{\psi} = 0$. This equation is linear in the activities, even though the activities can depend nonlinearly on the concentrations. Unlike probabilities, the activities are not normalized: their sum $S = \sum_n \psi_n$ can change with the rate constants. We define the normalized activity vector $\boldsymbol{\pi} = \boldsymbol{\psi} / S$.

Now change only the forward and backward rate constants r_+ and r_- of one reaction $j \rightleftharpoons i$. As in the derivation of Eq. (1), subtracting the two complex-balance equations gives

$$K \Delta \boldsymbol{\psi} = -\Delta K \boldsymbol{\psi}'. \quad (11)$$

Multiplying Eq. (11) by $G = K^D$, with $GK = I - \pi\mathbf{1}^\top$, and using $\mathbf{1}^\top \Delta\psi = \Delta S$ gives

$$\Delta\psi - \pi\Delta S = -G\Delta K\psi'. \quad (12)$$

For the controlled reaction, this yields

$$\begin{aligned} \Delta\psi_n &= \pi_n\Delta S + h_n\alpha, \\ \alpha &\equiv \Delta r_+\psi'_j - \Delta r_-\psi'_i, \end{aligned} \quad (13)$$

where $h_n = G_{nj} - G_{ni}$ and $\Delta S = S' - S$. The first term accounts for the change in the total activity; it vanishes for probabilities because normalization fixes $S = 1$. Equivalently, $\psi' = S'\pi + h\alpha$. Eliminating the two scalars S' and α between three complexes a, b, c gives

$$C_{bc}\psi'_a + C_{ca}\psi'_b + C_{ab}\psi'_c = 0, \quad (14)$$

where $C_{ab} = \pi_a h_b - \pi_b h_a$ is evaluated before the change and is independent of Δr_+ and Δr_- . Equation (14) recovers the homogeneous relation in Eq. (5) of Ref. [10].

V. CONCLUSION

We unify known mutual-linearity relations using the classical finite-response identity, Eq. (1). From this common starting point, we derive the relations for stationary probabilities, Eq. (3); state observables, Eq. (4); stationary currents, Eq. (8); components of linear systems, Eq. (9); Laplace-transformed probabilities, Eq. (10); and stationary complex activities in complex-balanced chemical reaction networks, Eq. (14).

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